



How To Make A BUSINESS CARD FLASHLIGHT

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The aim of this article is to make a business/visiting card useful, something that people really do not want to throw away and use as a piece of advertising. Essence of this project is that anyone can make it at home without any soldering, PCB and so on, as it requires none of these. The project can be extended to multiple interactive marketing media like posters. It only requires some basic materials as listed below and shown in Fig. 1:

1. Aluminium foil/aluminium tape/copper tape



Fig. 1: Materials and tools required for the project



Fig. 2: Tracing the circuit on the card

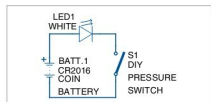


Fig. 3: Circuit diagram of the business card flashlight

2. 3mm white LED
 3. CR2016 button-cell battery
 4. Piece of foam or cardboard
 5. Two visiting/business cards
 6. Sheet of paper (optional)
 7. Pencil
 8. Adhesive (like Fevicol)
- Tools required:
1. Scissors
 2. Cutter
 3. Vernier calipers (optional)

Construction and testing

Step 1. Tracing the circuit. Take a lighter-grade pencil (3H/4H) and make a rough layout or path on one of your business cards along which the circuit will be placed (Fig. 2). Circuit diagram of the business card flashlight is shown in Fig. 3. Place the tape, LED and battery on the marked areas.



Fig. 4: Cutting the foam to make space for LED and battery



Fig. 5: Aluminium tape



Fig. 6: Placing aluminium tape and LED on the visiting card

The switch (explained in subsequent steps) will be at the location of the battery. Place the switch along the visual element of your business card, signifying a look of a button. Here, I have placed it along my company logo of a hand.

Interestingly, the circuit here does not need any resistance since we are using a white LED that has a typical forward voltage of 3.2 volts with a battery that has a maximum rated voltage of 3V. Current through the battery does not exceed the rated 25mA. Also, the battery has an internal resistance of 150-ohm, which is sufficient enough; hence, no external resistance is required.

In case you plan to use a red or green LED, consider using a resistor of 100- to 150-ohm.

Step 2. Cutting the cardboard/foam. The cardboard/foam is used to provide spacing between the two visiting cards. That is, the foam will be sandwiched between the two cards. It also makes a modular casing for LED and battery (Fig. 4).

First, cut the foam/cardboard as per the dimensions of the visiting card. Then, cut small openings for placing the LED and battery. I prefer foam because it adds negligible weight, and is available in various colours.

Thickness of the foam should be greater than that of battery. You may need Vernier calipers to measure that but a standard 3mm to 4mm foam will work. Table 1 shows the thickness of various button cells—we choose CR2016 because it has minimum thickness.

Step 3. Making an aluminium